

FORMULA & FUNCTION SYNTAX

Function Syntax

Reference Types

Error Types

Counting Functions

Logical Functions

Lookup Functions

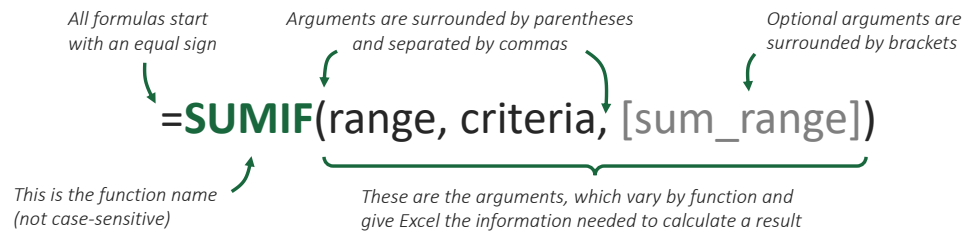
Date & Time Functions

Text Functions

Formulas are calculations performed in cells that start with an equal sign “=”

Functions are preset calculations used in formulas that take arguments as inputs between parentheses and return a corresponding output

- **NOTE:** Formulas can have multiple functions, or none at all ($= A1 + B1$ is a valid formula)



PRO TIP: As you begin writing a function, an “IntelliSense” window pops up with details on the function syntax and arguments

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TYPES OF FUNCTIONS

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There are three **types of functions**:

1 Traditional functions that return **one value**

=SUM(D2:D13)			
C	D	E	F
Month	Flights		Total
Jan	389		4,985
Feb	356		
Mar	405		
Apr	391		
May	448		
Jun	417		
Jul	476		
Aug	457		
Sep	437		
Oct	405		
Nov	390		
Dec	414		

2 Array functions that return **multiple values**

=UNIQUE(C2:C9)		
C	D	E
Airline		Unique List
Southwest		Southwest
Delta		Delta
United		United
United		American
American		
Southwest		
Delta		
American		

The formula only lives in the first cell in the range

3 **Volatile** functions that recalculate automatically

=NOW()	
C	
Current Date & Time	
3/5/2025 22:33:37	

These can either return one value or multiple

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COMMON ERROR TYPES

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Error type	What it means	How to fix it
#NAME?	The function name isn't recognized	Make sure that function names are correct, references are valid, and quotation marks and colons are in place
#VALUE!	Function has the wrong type of argument	Check that you're not trying to perform an arithmetic operation on text strings or cells formatted as text
#DIV/0!	Formula is dividing by zero or an empty cell	Check the value of your divisor; if 0 is correct, use an IF statement to display an alternate value if you choose
#REF!	Formula refers to a cell that is not valid	Make sure that you didn't move, delete, or replace cells that are referenced in your formula
#N/A!	A lookup function didn't find a match	Check the lookup keys and ranges; if no match is correct, use an IF statement to display an alternate value
#SPILL!	Something prevents a range from spilling	Check if there is any text or merged cells obstructing the range from spilling (you also can't spill in an Excel table)

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COUNTING FUNCTIONS

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Counting functions are used to count the cells in a range in different ways:

COUNT()	Counts the numeric values in a range	=COUNT (value1, [value2, ...])
COUNTA()	Counts the non-empty cells in a range	=COUNTA (value1, [value2, ...])
COUNTBLANK()	Counts the empty cells in a range	=COUNTBLANK (value1, [value2, ...])

This accepts cell ranges 

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LOGICAL FUNCTIONS

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Logical functions are Excel's *decision-making* tools

- **IF** a logical expression is true **THEN** do this, **OTHERWISE** do that
- A logical expression compares numbers, dates, text, cells or functions using **operators**

Operator	Name	Example	Result
=	Equal to	10 = 7	FALSE
>	Greater than	10 > 7	TRUE
<	Less Than	"e" < "r"	TRUE
>=	Greater than or equal to	"e" >= "r"	FALSE
<=	Less than or equal to	2012-05-19 <= 2008-05-21	FALSE
<>	Not equal to	2012-05-19 <> 2008-05-21	TRUE



HEY THIS IS IMPORTANT!

Excel interprets the logical value of **TRUE** as **1**, and the logical value of **FALSE** as **0**

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These are the most common **logical functions** used:

IF()

Performs a logical test and returns a value if TRUE, or another if FALSE

=IF(logical, [if_true], [if_false])

AND()

Performs one or more logical tests and returns a value of TRUE if all arguments are true

=AND(logical1, [logical2] , ...)

OR()

Performs one or more logical tests and returns a value of TRUE if any argument is true

=OR(logical1, [logical2] , ...)

IFERROR()

Evaluates a formula and returns an alternative value if the formula returns an error

=IFERROR(value, if_error)

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